

Mini Project 1: “Student Productivity Tracker” Objective: Combine loops, arrays, and file handling to build a console-based app that logs daily study hours and generates a weekly progress report. Trainer Coverage: Guide students to design menu-driven interface. Discuss file I/O for persistence.

Tasks

1.

Create a console-based app called Music Listening Logger that allows users to enter the number of minutes they listened to music each day for a week and stores this data in an array.

2.

Add a menu-driven interface to your Music Listening Logger so users can choose to log new listening minutes, view their weekly summary, or exit the app.

Hint: Use a loop to repeatedly show the menu until the user selects exit.

3.

Implement file handling in your Music Listening Logger so that when the user logs their daily listening minutes, the data is saved to a file named music_log.txt for persistence.

4.

Add a feature to your app that reads the saved music_log.txt file and generates a weekly report showing the total, average, and highest listening minutes in the week.

5.

Refactor your Music Listening Logger to allow users to reset their weekly data by clearing the array and deleting the contents of music_log.txt.

Constraint: Make sure the reset option is available in the menu and asks for confirmation before deleting data.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#define DAYS 7
```

```
// Function declarations
```

```
void logMusic(int minutes[]);
void viewSummary(int minutes[]);
void saveToFile(int minutes[]);
void weeklyReport();
void resetData(int minutes[]);

int main()
{
    int minutes[DAYS] = {0};
    int choice;

    do
    {
        printf("\n===== MUSIC LISTENING LOGGER =====\n");
        printf("1. Log daily listening minutes\n");
        printf("2. View weekly summary\n");
        printf("3. Generate weekly report\n");
        printf("4. Reset weekly data\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                logMusic(minutes);
                saveToFile(minutes);
                break;
```

```
case 2:
    viewSummary(minutes);
    break;

case 3:
    weeklyReport();
    break;

case 4:
    resetData(minutes);
    break;

case 5:
    printf("\nThank you for using Music Listening Logger!\n");
    break;

default:
    printf("\nInvalid choice. Please try again.\n");
}

} while (choice != 5);

return 0;
}

// Task 1: Enter music listening minutes
```

```

void logMusic(int minutes[])
{
    char *days[] = {
        "Monday",
        "Tuesday",
        "Wednesday",
        "Thursday",
        "Friday",
        "Saturday",
        "Sunday"
    };

    printf("\nEnter music listening minutes:\n");

    for (int i = 0; i < DAYS; i++)
    {
        printf("%s: ", days[i]);
        scanf("%d", &minutes[i]);
    }

    printf("\nWeekly listening data saved in array.\n");
}

```

// Task 2: Display weekly summary

```

void viewSummary(int minutes[])
{
    char *days[] = {

```

```
"Monday",  
"Tuesday",  
"Wednesday",  
"Thursday",  
"Friday",  
"Saturday",  
"Sunday"  
};
```

```
printf("\n===== WEEKLY SUMMARY =====\n");
```

```
for (int i = 0; i < DAYS; i++)  
{  
    printf("%s: %d minutes\n", days[i], minutes[i]);  
}  
}
```

```
// Task 3: Save data into file
```

```
void saveToFile(int minutes[])
```

```
{  
    FILE *file;  
  
    file = fopen("music_log.txt", "w");  
  
    if (file == NULL)  
    {  
        printf("Error opening file!\n");  
    }  
}
```

```
        return;
    }

    for (int i = 0; i < DAYS; i++)
    {
        fprintf(file, "%d\n", minutes[i]);
    }

    fclose(file);

    printf("Data saved successfully to music_log.txt\n");
}
```

// Task 4: Read file and generate weekly report

```
void weeklyReport()
{
    FILE *file;
    int minutes[DAYS];
    int total = 0;
    int highest = 0;
    int count = 0;

    file = fopen("music_log.txt", "r");

    if (file == NULL)
    {
        printf("\nNo saved music data found.\n");
    }
}
```

```

        return;
    }

    // Read data from the file
    while (count < DAYS && fscanf(file, "%d", &minutes[count]) == 1)
    {
        count++;
    }

    fclose(file);

    if (count == 0)
    {
        printf("\nNo data available.\n");
        return;
    }

    // Calculate total and highest listening time
    for (int i = 0; i < count; i++)
    {
        total += minutes[i];

        if (minutes[i] > highest)
        {
            highest = minutes[i];
        }
    }
}

```

```
float average = (float)total / count;
```

```
printf("\n===== WEEKLY REPORT =====\n");  
printf("Total listening time : %d minutes\n", total);  
printf("Average listening time : %.2f minutes\n", average);  
printf("Highest listening time : %d minutes\n", highest);  
}
```

```
// Task 5: Reset data
```

```
void resetData(int minutes[])
```

```
{
```

```
    char confirmation;
```

```
    printf("\nAre you sure you want to delete weekly data? (y/n): ");
```

```
    scanf(" %c", &confirmation);
```

```
    if (confirmation == 'y' || confirmation == 'Y')
```

```
    {
```

```
        // Clear array
```

```
        for (int i = 0; i < DAYS; i++)
```

```
        {
```

```
            minutes[i] = 0;
```

```
        }
```

```
        // Clear file contents
```

```
        FILE *file = fopen("music_log.txt", "w");
```

```
if (file != NULL)
{
    fclose(file);
}

printf("\nWeekly data has been reset successfully.\n");
}
else
{
    printf("\nReset cancelled.\n");
}
}
```



C:\Users\gajja\OneDrive\Desk



===== MUSIC LISTENING LOGGER =====

1. Log daily listening minutes
2. View weekly summary
3. Generate weekly report
4. Reset weekly data
5. Exit

Enter your choice: 1

Enter music listening minutes:

Monday: 10
Tuesday: 10
Wednesday: 10
Thursday: 20
Friday: 20
Saturday: 30
Sunday: 60

Weekly listening data saved in array.
Data saved successfully to music_log.txt

===== MUSIC LISTENING LOGGER =====

1. Log daily listening minutes
2. View weekly summary
3. Generate weekly report
4. Reset weekly data
5. Exit

Enter your choice: 2

===== WEEKLY SUMMARY =====

Monday: 10 minutes
Tuesday: 10 minutes
Wednesday: 10 minutes
Thursday: 20 minutes
Friday: 20 minutes
Saturday: 30 minutes
Sunday: 60 minutes

===== MUSIC LISTENING LOGGER =====

1. Log daily listening minutes
2. View weekly summary
3. Generate weekly report
4. Reset weekly data
5. Exit

Enter your choice: 3

===== WEEKLY REPORT =====

Total listening time : 160 minutes
Average listening time : 22.86 minutes
Highest listening time : 60 minutes

===== MUSIC LISTENING LOGGER =====

1. Log daily listening minutes
2. View weekly summary
3. Generate weekly report
4. Reset weekly data
5. Exit

Enter your choice: 5

Thank you for using Music Listening Logger!

Process exited after 48.27 seconds with return value 0
Press any key to continue . . . |